

WASP-11b

R-1667

Benchmark run · workflow W8-benchmark-deep · record deep-bench_HATP-10_210907 · created 2026-08-02T10:10:13+00:00

BENCHMARK: NON-DETECTION

Results

Mid-Transit Time (Tmid)	2459464.8972 +/- 0.0028 BJD_TDB
Ratio of Planet to Stellar Radius (Rp/R*)	0.086 +/- 0.042
Transit depth (Rp/Rs)^2	0.74 +/- 0.73 [%]
Orbital Inclination (inc)	86.2 +/- 1.67
Airmass coefficient 1 (a1)	1.194 +/- 0.019
Airmass coefficient 2 (a2)	-0.1518 +/- 0.0097
Scatter in the residuals of the lightcurve fit is	1.51 %
Best Comparison Star	None
Optimal Aperture	5.54
Optimal Annulus	5.54
Transit Duration (day)	0.059 +/- 0.035

Accuracy vs published ephemeris (ExoClock/NEA)

Rp/R■ fit vs published	0.086 ± 0.042 vs 0.131 (deviation 0.9σ)
Duration fit vs published	0.059 d vs 0.1075 d (deviation 1.16σ)
Mid-time O–C	-5.6 min
Depth SNR	1.7
Residual scatter	1.51 %

Light curve

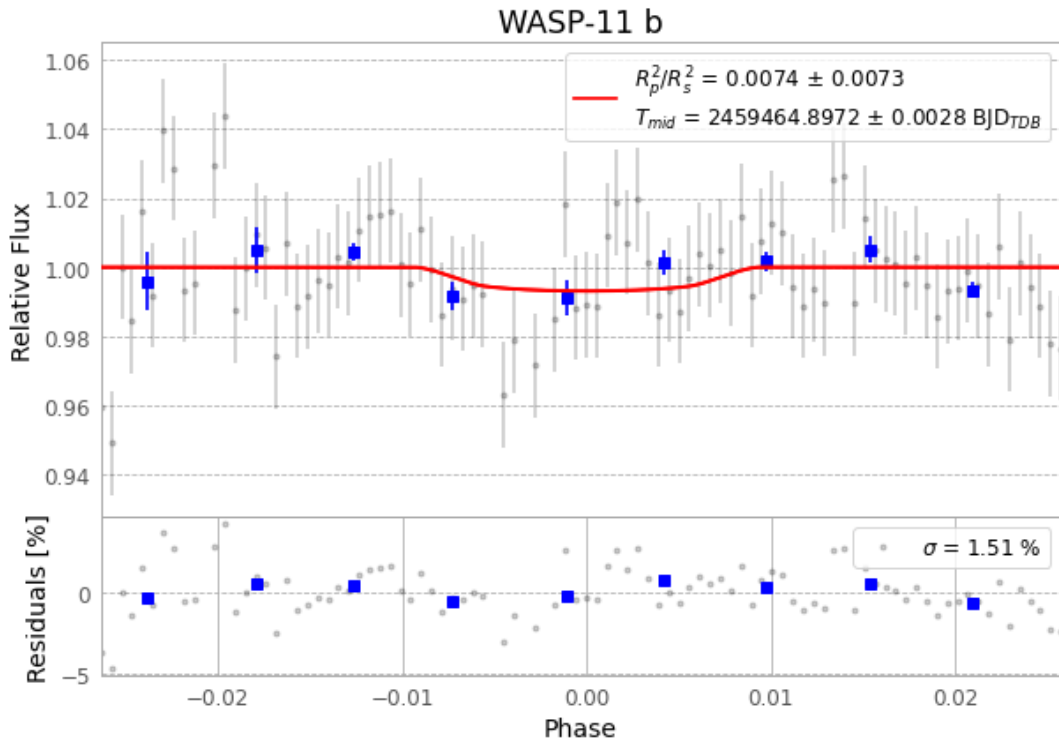


Figure 1 — FinalLightCurve_WASP-11 b_2021-09-07.png (SHA-256 7f0cb76d1253c0bc...)

Method & software

EXOTIC 4.3.1 aperture photometry with nested-sampling transit fit (ultranest 3.6.5); priors from the NASA Exoplanet Archive. Camera CCD, binning 1x1, filter CV, target pixel [363, 337], 3 comparison star(s). Exact configuration: parameters.inits_json in record.json (SHA-256 601e4b6ddd6b0ce7...).

exotic 4.3.1 · astropy 6.1.7 · photutils 2.0.2 · numpy 1.26.4 · scipy 1.14.1 · twirl 0.5.1 · astroquery 0.4.11 · ultranest 3.6.5 · python 3.12.13 · pipeline_commit ba3ef8b

Data provenance

94 FITS frames (62 MB), HATP-10210907070612.FITS → HATP-10210907114512.FITS. Every input frame is SHA-256-fingerprinted in record.json; raw frames available on request and verifiable against that manifest. Artifacts: bench-HATP-10-210907.log (8b27ca943c67...), FinalLightCurve_WASP-11 b_2021-09-07.png (7f0cb76d1253...), FinalLightCurve_WASP-11 b_2021-09-07.csv (d539a6ade9b0...)

Compute resources

Wall time 130.2 s · peak memory None MB · host mac-mini-m1-8gb (macOS-26.4-arm64-arm-64bit)

Observer: — · AAVSO obscode ABGA · Automated pipeline with human review — nothing is submitted without a human approving this specific result. Data license CC-BY-4.0. Generated 2026-08-02 10:10Z.